

TASK 3

Sensor Data Logger Using SD Card

Temperature & Light Data Collection with Timestamped CSV Storage

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Task: Collect sensor data (e.g., temperature and light) and save timestamped entries to an SD card (CSV).

1. Objective

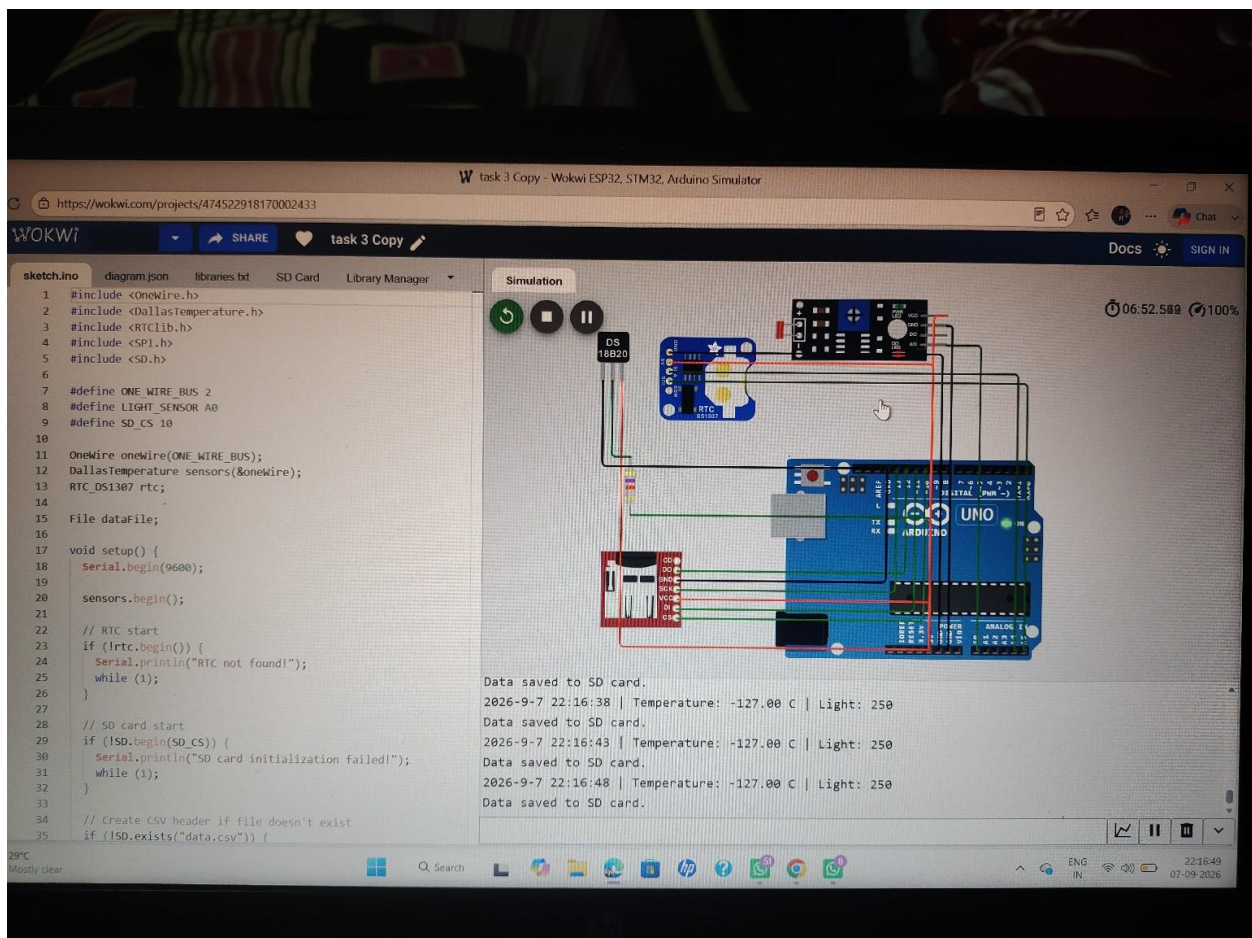
To collect temperature and light sensor readings, add a real-time timestamp using an RTC module, and save the readings on an SD card in CSV format for monitoring and later analysis.

2. Components Used

- Arduino UNO
- DS18B20 temperature sensor
- LDR/light sensor module
- DS1307 RTC module
- MicroSD card module and SD card
- Resistor, breadboard and jumper wires

3. Circuit, Code & Simulation Evidence

The uploaded project screenshot shows the Arduino UNO connected to the DS18B20 temperature sensor, RTC module, light sensor and SD card module. It also shows the Arduino code and the running simulation output.



4. Pin Connections

Component	Signal / Pin	Arduino Pin
DS18B20	Data	D2
Light Sensor	Analog Output	A0
DS1307 RTC	SDA	A4
DS1307 RTC	SCL	A5
SD Card	CS	D10
SD Card	MOSI	D11
SD Card	MISO	D12
SD Card	SCK	D13
Modules	VCC / GND	5V / GND

5. Program / Code

The project uses OneWire/DallasTemperature for the DS18B20, RTCLib for the RTC, and SPI and SD libraries for SD-card communication. The screenshot contains the working source code used in the simulation.

6. Timestamped CSV Data

The simulation output demonstrates repeated entries being saved to the SD card. The intended CSV structure is:

Timestamp	Temperature	Light
2026-09-07 22:16:38	-127.00 °C	250
2026-09-07 22:16:43	-127.00 °C	250
2026-09-07 22:16:48	-127.00 °C	250

Note: The screenshot visibly reports the temperature as -127.00 °C. This commonly indicates that the DS18B20 reading was not detected correctly in the simulation; the evidence is retained as shown rather than altered.

7. Project Links

Wokwi Project Link: <https://wokwi.com/projects/474522918170002433>

Google Drive Link:

https://drive.google.com/drive/folders/11dqLF0Z8lDaWmx2JauwXJU1GQkmTAcPL?usp=drive_link

8. Conclusion

The Task 3 prototype demonstrates a sensor data logging system in which temperature and light readings are timestamped with an RTC and stored on an SD card for CSV-based data logging. The supplied screenshot provides evidence of the circuit, source code and successful SD-card write messages during simulation.